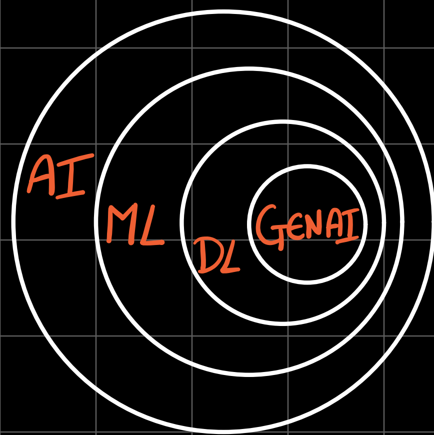
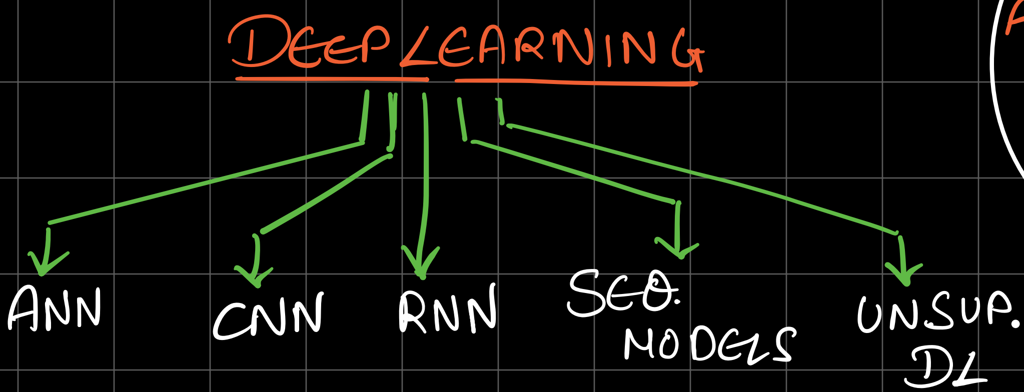
GEN-AI

It is a type of AI that creates new content by learning from existing data, mimicking human creativity.





SEQUENCE TASKS :

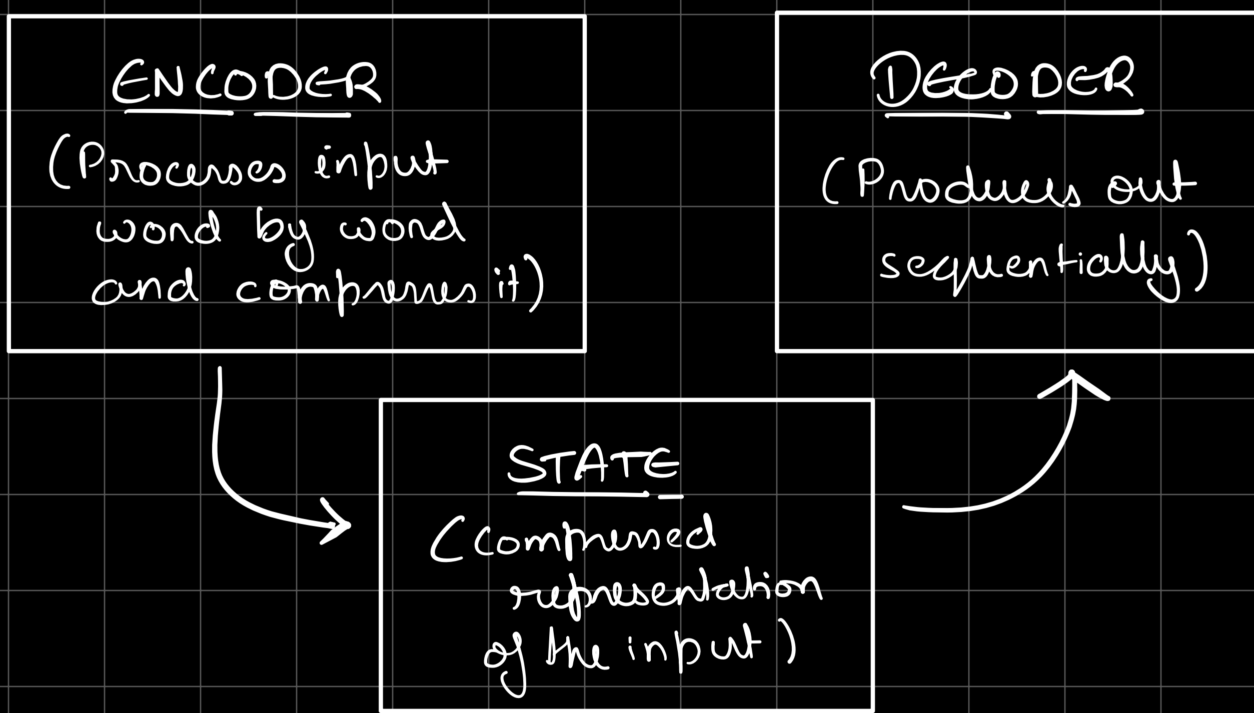
1) MANY TO ONE (Sentiment analysis)
2) ONE TO MANY (Image discription)
3) SYNCHRONOS SEQ TO SEQ (Parts of speech recog)
4) ASYNCHRONOS SEQ. TO SEQ. (Translation)

WAYS TO SOLVE SEQ TO SEQ :

1) ENCODER-DECODER
2) ATTENTION MECHANISM
3) TRANSFORMER
4) TRANSFER LEARNING

5) LLM's

ENCODER-DECODER



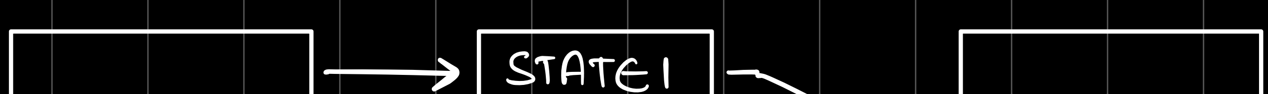
⇒ Worked good for small input.

⇒ Problems with memory loss.

ATTENTION MECHANISM :

⇒ Has a similar encoder block but the decoder block is different.

⇒ All intermediate states are available unlike encoder-decoder which only has final state.



Attention layer is a neural network that finds the most relevent state for the given output of the decoder.

⇒ The traning time / computational complexity of the attention mechanism is really high.

TRANSFORMER:

⇒ It still has encoder & decoder.

⇒ It can read all of inputs content all at once (Parallel Processing).

⇒ It can be trained in fraction of time as compared to previous arch.

⇒ The only drawback was that it needs a lot of data and it cannot be trained on small amounts of data.

TRANSFER LEARNING:

⇒ In this knowledge learnt from a previous task can be used to boost performance on a related task.

LLM's:

⇒ Trained on huge datasets which made them extremely efficient for any task.

⇒ Hardware heavy

⇒ Works on transformer model

TRANING:

① UNSUPERVISED GENRATIVE PRETRAINING:

⇒ They are fed large amounts of data that they start to recognize and predict patterns and learn from.

⇒ BACK-PROPAGATION: Input with known outputs is fed to the model and the parameters are tweaked till they start predicting the desired output.

② REINFORCEMENT LEARNING BY HUMAN FEEDBACK:

Humans test and evaluate the output by the model and give feedback.

Continual Learning Models suffer from catastrophic forgetting. This basically means when the model learns new information it forgets / overwrites the old weights.